고려대학교 약학대학
KOREA UNIVERSITY COLLEGE OF PHARMACY

KU-The Future
110 years and beyond

Microbiology



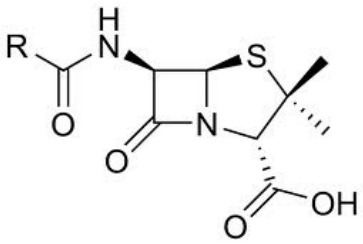
Fall semester, 2026

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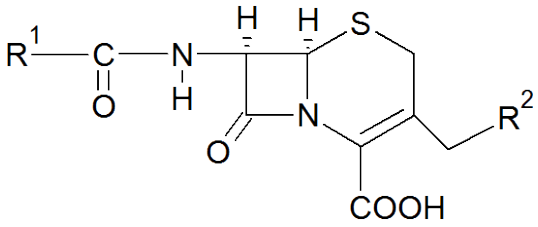
❖ Non-classical beta-lactams

▶ Carbapenems

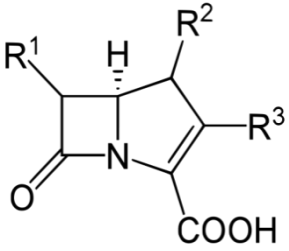
- a class of β-lactam antibiotics: possessing a β-lactam ring and sharing the same mechanism of action of β-lactams
- be structurally unique: differ from both penicillins and cephalosporins → β-lactam + pyrroline (not having a sulfur)
- highly resistant to most β-lactamases
- initially isolated from actinomycetes (방선균; genus *Streptomyces*)



Penicillin core structure



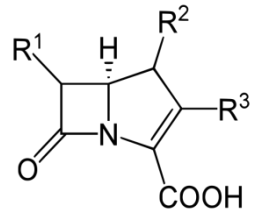
Core structure of cephalosporins
Basic structure (1st)



Carbapenem core structure

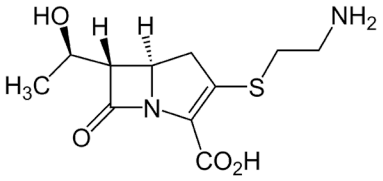
Carbapenems

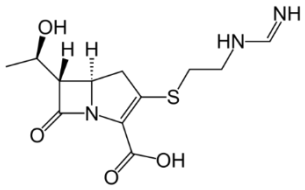
- be among our most broad-spectrum antibacterial drugs (having a broad spectrum of antibacterial activity)
  - due to their compact structures → easily pass through the porin
  - good activity: both G (+) and G (−) bacteria
    : including *Pseudomonas, enterococci,* enteric GNRs
  - poor activity: MRSA, penicillin-resistant streptococci
- natural carbapenem: **thienamycin** (from *Streptomyces cattleya*)
- semisynthetic carbapenems: **imipenem**/cilastatin, **meropenem**, ertapenem, doripenem, panipenem, biapenem, etc.
  - imipenem: can be hydrolyzed in a kidney by dehydropeptidase to a nephrotoxic metabolite
  - cilastatin: dehydropeptidase inhibitor / co-administered with imipenem



Carbapenem core structure

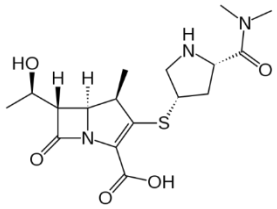
Thienamycin

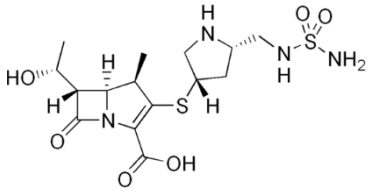




Imipenem

Meropenem





Doripenem

▶ Carbapenems

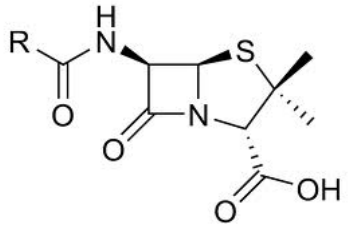
- ertapenem: not active against *Pseudomonas* and *enterococci*
  - poor choice for many nosocomical infections, but more convenient than the others because it is administered only once a day.
  - pain when be administered intramuscularly → together with 1% lidocaine

· Adverse effects:
- be similar to those of other β-lactams (nausea, vomiting, etc.) / seizures (in patients who have renal dysfunction)
- induce cross-hypersensitivity with penicillins (1-50%)

· Resistance:
- currently, uncommon, but has been raised
- provided by New Delhi metallo-β-lactamase (NDM-1)

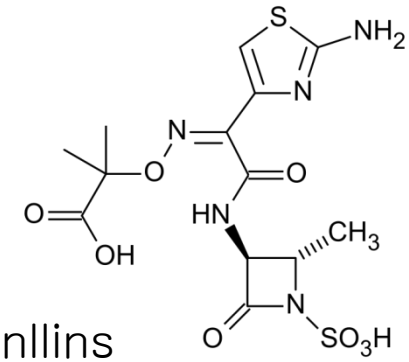
❖ Non-classical beta-lactams

▶ Monobactams (단핵베타락탐류)



Penicillin
core structure

- A class of β-lactam antibiotics: possessing only the 4-membered ring of the basic β-lactam structure (the β-lactam ring is alone and not fused to another ring.)
  → hence the name monobactam
- Members:
  - natural: sulfazecin (설파제신) and nocardicin (노카르디신)
    - sulfazecin: produced from G (-) bacteria (e.g. *Chromobacterium*, *Agrobacterium*)
    - nocardicin: produced from G (+) bacteria (e.g. *Nocardia uniformis*)
  - semisynthetic: **aztreonam** (아즈트레오남)
- Mode of action: similar as the other β-lactams
- Spectrum of activity: G (-) only; poor to G (+)
  - active against *Pseudomonas* and most GNRs
    : strongly bind to PBP3
- Low toxicity; not induce cross-hypersensitivity with penicinllins
  → azthreonam: nausea, vomiting, rash / rarely toxic epidermal necrolysis (독성 표피 괴사증)



Aztronam

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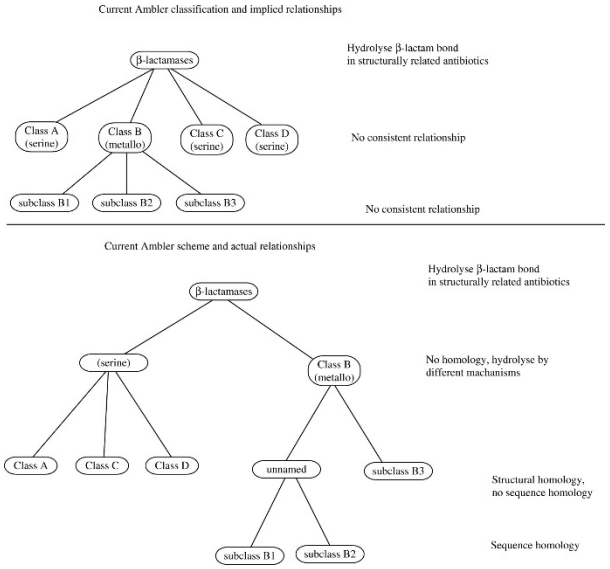
❖ Two mechanisms of resistance to the β-lactams

1. Production of β-lactamase

- be produced by some bacteria (>200 species)
- be responsible to β-lactam antibiotics (penicillins, cephems, and carbapenems)
- classifications:

1) molecular classification (Ambler classification):

- based on the amino acid sequences
- four classes: A to D
  - class A, C, and D: active-site serine-based β-lactamases (serine-based mechanisms; 세린형)
  - class B: metallo-β-lactamases (zinc ion (Zn^{2+})-based mechanisms; 금속형)



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1. production of β-lactamase

2) functional classification (Buch classification):

: based on their functions / four groups: 1 to 4

· group 1: cephalosporinases not inhibited by clavulanic acid (mole. C)

· group 2: penicillinases, cephalosporinases, or both inhibited by clavulanic acid (mostly mole. A)

- 2a: penicillinase (inactivating penicillins)
- 2b: broad-spectrum; inactivating both penicillins and cephalosporins
- 2be: extended spectrum; inactivating the 3rd cephalosporins and monobactams
- 2br: inhibitor-resistant; inactivating both penicillins and cephalosporins, but reduced binding to clavulanic acid
- 2c: carbenicillinase (inactivating carbenicillins)
- 2d: cloxacilanase (iniactivating cloxacillin) / reduced binding to clavulanic acid (mole. D or A)
- 2e: cephalosporinases
- 2f: carbapenamases (serine-based mechanism)

· group 3: carbapenamases (zinc-based mechanism) (mole. B)

· group 4: penicillinases not inhibited by clavulanic acid

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❖ Two mechanisms of resistance to the β-lactams

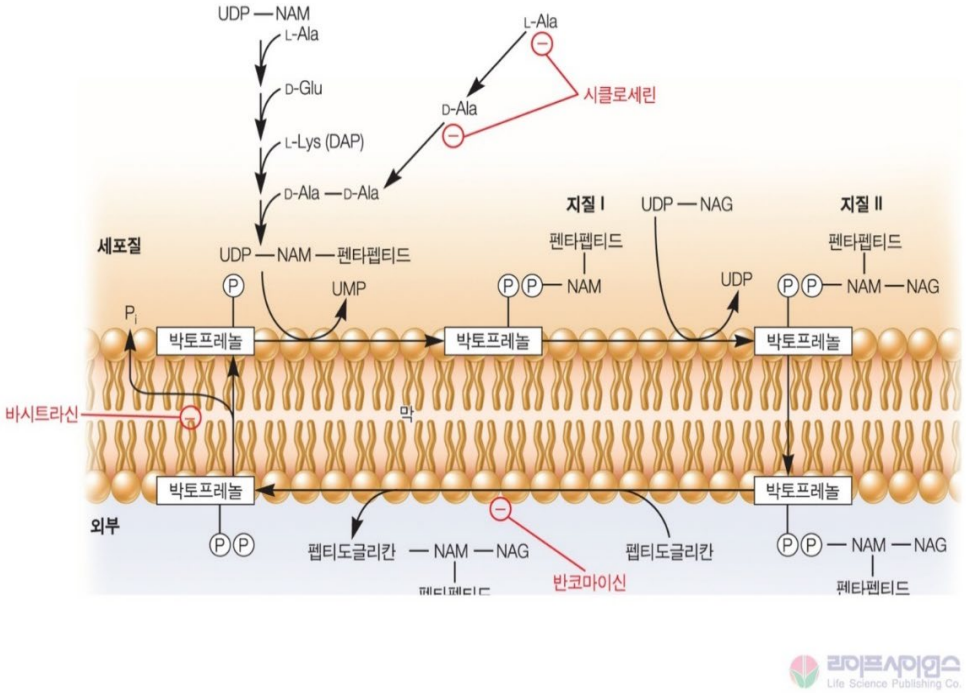
1. Production of β-lactamase

2. Modification of penicillin-binding proteins (PBPs)

- proteins having the affinity for and binding to penicillin
- all β-lactams can bind to PBPs (with an exception)
- different PBPs exist in different bacteria.
- modification:
  - formation of new PBPs
    ex) PBP2a in MRSA (vs. MSSA: PBP1, 2, 3 and 4)
  - formation of PBPs having low affinity for antibiotics
    ex) PSSP and PRSP: both have 6 PBPs,
    but some of PBPs in PRSP have low affinity for penicillin

Non-beta-lactam cell wall synthesis inhibitors

❖ Other antibacterial agents that act on cell walls

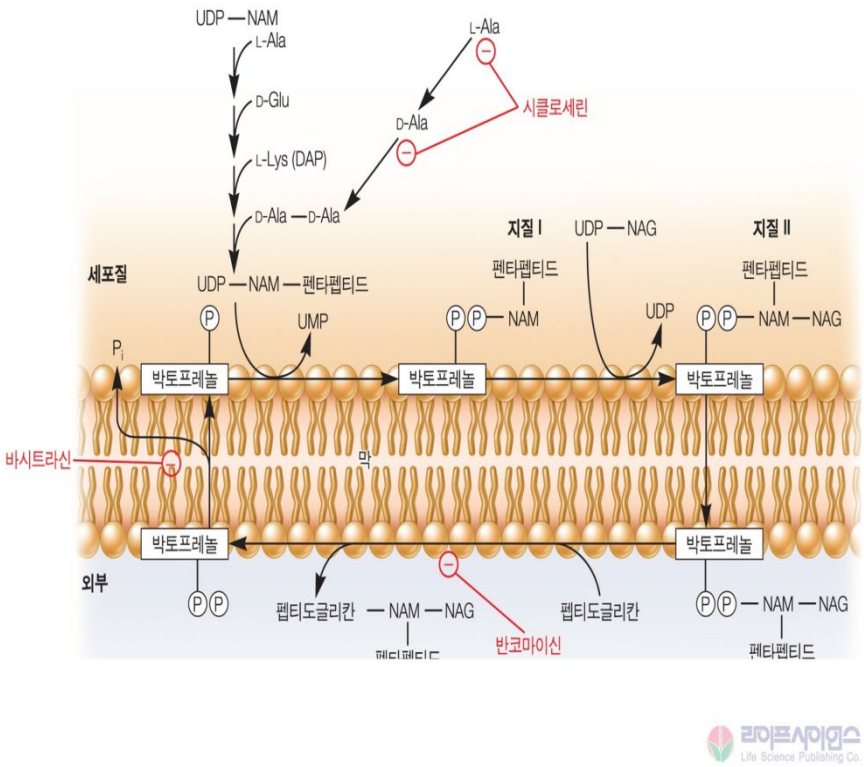


∗ Steps of peptidoglycan synthesis

1. Synthesis of peptidoglycan monomers in the cytosol
2. Attachment of the monomer to a membrane carrier (called bactoprenol)
3. Transportation of the monomers across the cytoplasmic membrane (by the bactoprenol)
4. Insertion of the monomers into the grown peptidoglycan chains (by the bactoprenol)
5. Lose of a phosphate group on the way back to the cytoplasmic membrane
6. Bactorprenol to be recycled and to pick up another monomer

Non-beta-lactam cell wall synthesis inhibitors

❖ Other antibacterial agents that act on cell walls



▶ cycloserine (Seromycin):

– competitively inhibiting an alanine racemase and D-alanine ligase
· alanine racemase: L- ↔ D-form

– fosfomycin

▶ vancomycin (Vancocin):

– inhibiting the cross-linking of the peptidoglycan monomers by binding with two D-alanine residues

– teichoplanin, ristocetin, moenomycin

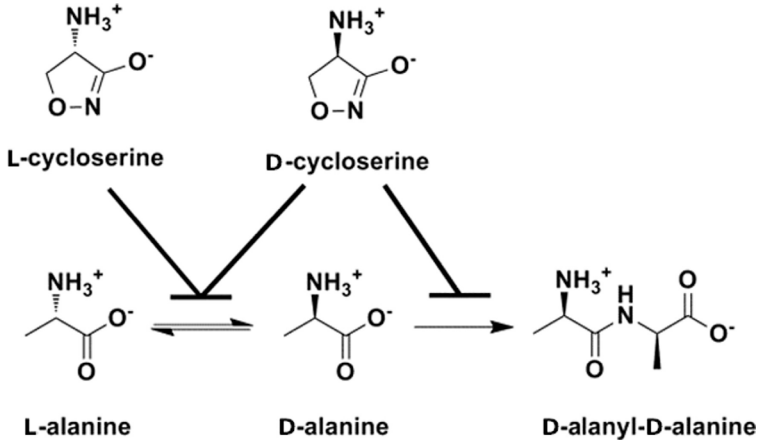
▶ bacitracin (Baciim):

– interfering with the dephosphorylation of the bactoprenol phosphate

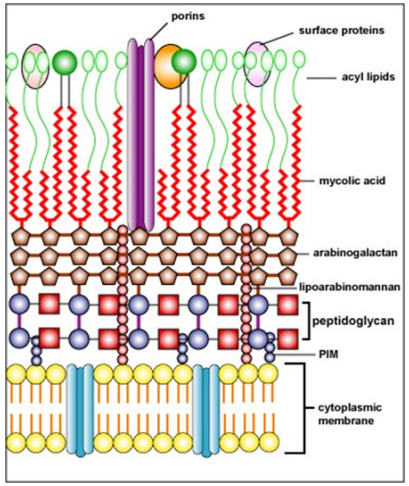
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❖ Other antibacterial agents that act on cell walls

▶ Cycloserine



- an analog of the amino acid alanine / discovered from the genus *Streptomyces*
- a broad-spectrum antibiotic, but specifically used in tuberculosis, caused by *Mycobacterium tuberculosis* (not commonly used for bacterial infections other than tuberculosis)
  → a second-line drug for tuberculosis (only considered if one or more first-line drugs cannot be used; drug-resistant *M. tuberculosis*)
- **adverse effects**: CNS manifestations (neurotoxicity; headache, depression, psychosis convulsions, etc.)
- *M. tuberculosis*: phospholipids and mycolic acids of the outer membrane → make a waxy layer: resisting penetration from antibiotics
- **mechanism of action**: competitively inhibiting alanine racemase and D-Ala ligase (D-alanylalanine synthetase)



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❖ Other antibacterial agents that act on cell walls

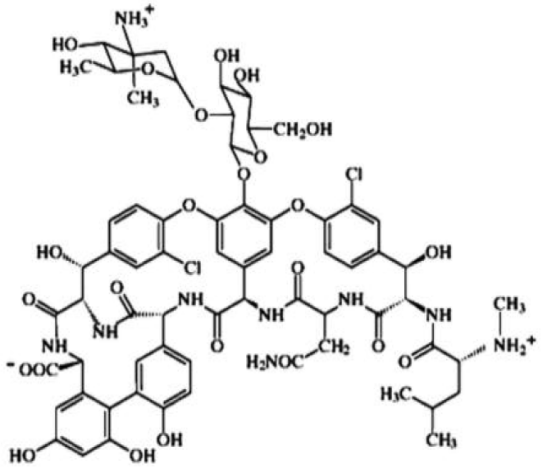
▶ Vancomycin ("vanquish"; against PRSAs):

· General aspects:

- "vanquish" against PRSAs
- a large, complex glycopeptide (tricyclic glycopeptide)
- an actinomycete, *Streptomyces orientalis*
- little absorbed by the gastrointestinal tract → intravenously

· Spectrum of activity:

- **effective for G (+)**, NOT for G (−) bacteria
  (too large to pass through porins in the outer membrane of G (−) bacteria)
- generally prescribed:
  · to combat severe infections of G (+) bacteria, resistant to other antibiotics (e.g., MRSA)
  · for patients being allergic to the classical beta-lactams



❖ Other antibacterial agents that act on cell walls

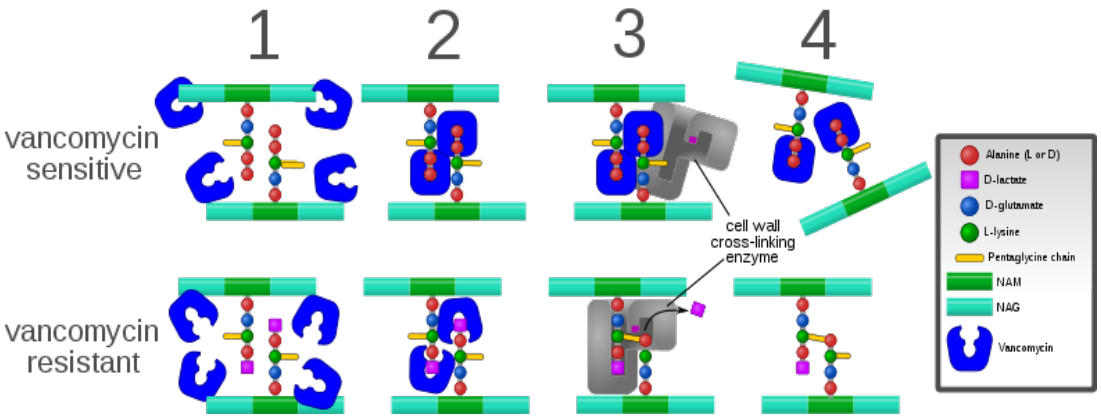
▶ Vancomycin ("vanquish"; against PRSAs):

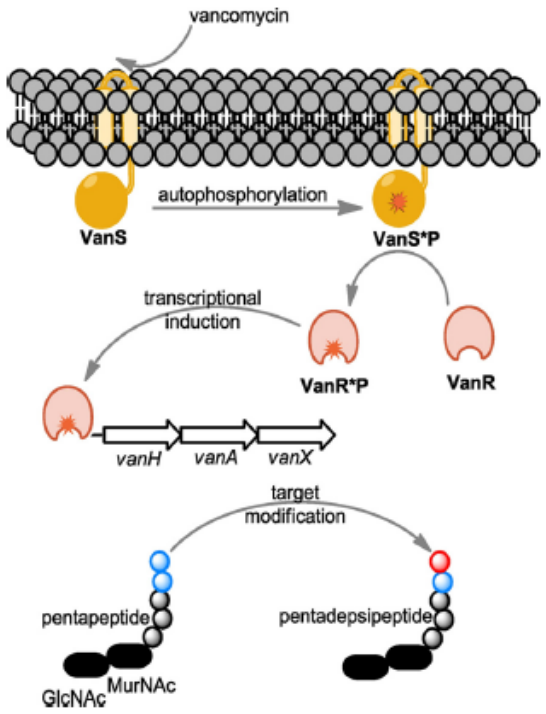
· Mode of action:

- binding with the two D-Ala residues (D-Ala-D-Ala)
  → inhibiting the interaction b/w the PBP and the two D-Ala residues
  → preventing the cross-linking of the peptidoglycan chains

· Mechanism of resistance:

- ex) vancomycin-resistant enterococci (VRE)
- producing the D-Ala-D-Ala-lactate (by the *VanHAX* operon)

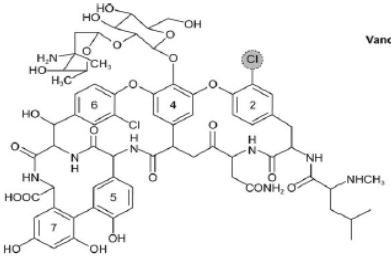


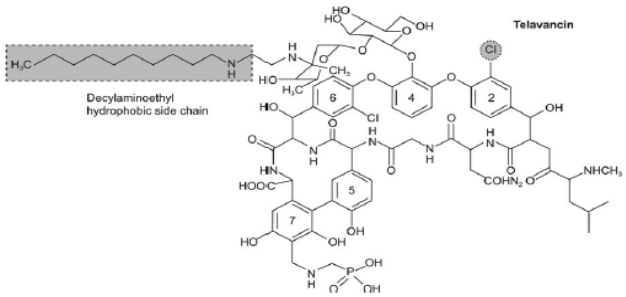


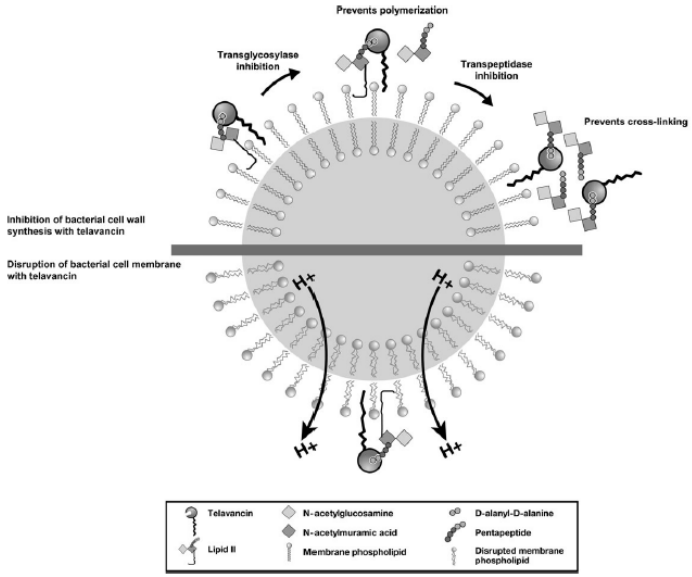
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Vancomycin ("vanquish"):

- Telavancin (trade name vibativ):
  - a lipoglycopeptide / semi-synthetic derivative of vancomycin
  - **improved activity** than vancomycin
    → effective against MRSA that is less susceptible to vancomycin
    (e.g., vancomycin-intermediate/resistant MRSA)
  - **dual mechanism** of antibacterial action:
    - binding with the two D-Ala residues
      → preventing the cross-linking of the peptidoglycan chains
    - binding with bacterial cell membranes (including cell wall precursor lipid II; bactoprenol-NAM-NAG)
      → depolarization and increased permeability of membrane







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❖ Other antibacterial agents that act on cell walls

▶ Bacitracin

- a polypeptide antibiotic: a mixture of related cyclic peptides (A, A1, B, B1, B2, C, D, E, F, G and X)
- produced from *Bacillus subtilis var Tracy*
- **spectrum of activity**: G (+) bacteria (including MRSA)
- **adverse effects**: potential risk of nephrotoxicity and life-threatening allergy reactions (anaphylaxis)
  when used internally
  - primarily used as a topical preparation (skin and eye ointments or drops)
  - (vs.) injection products authorized in Canada
    : ex) infants with pneumonia caused by *S. aureus*
- **mode of action**: interfering with the dephosphorylation of the bactopronol phosphate

